

Walton Plaza - Senior Frontend Developer Evaluation Task

Objective

Design and implement a high-performance product listing and detail system using Next.js, React 19, TypeScript, Tailwind CSS, and GraphQL.

Tech Stack

Next.js (App Router)
TypeScript (strict mode)
Tailwind CSS
GraphQL (Apollo/urql/custom)
React 19

Section 1: Architecture & Setup

1. Setup Next.js App Router with TypeScript strict mode
2. Define scalable folder structure
3. Configure GraphQL client with caching
4. Typed GraphQL queries
5. Environment-based API config

Section 2: Product Listing Page (PLP)

6. Fetch products via GraphQL
7. Pagination or infinite scroll (justify choice)
8. Loading skeleton & error handling
9. Filters (price, category, availability)
10. Sorting (price, rating)

Section 3: Product Card

11. Reusable ProductCard component
 - Optimized images

- Hover micro-interaction
- Optimistic add to cart

Section 4: Product Details Page (PDP)

- 12. Dynamic routing
- 13. Fetch product details
 - Image gallery
 - Variant selection
 - Stock-aware CTA
 - Dynamic pricing

Section 5: State Management

- 14. Cart system (add/remove/update)
- 15. State persistence and explanation of approach

Section 6: Performance Optimization

- 16. Memoization strategy
- 17. Server vs Client components
- 18. GraphQL optimization

Section 7: React 19

- 19. Use at least one modern React 19 feature

Deliverables

GitHub repository
README with architecture decisions and trade-offs

Evaluation Criteria

Architecture quality
Performance optimization
Code quality
TypeScript usage
Decision making ability